AI-Powered Credit Risk Intelligence Platform

NeoStats AI Engineer Internship — Candidate Assignment

Home Credit Default Risk dataset (307,511 applicants, real Kaggle data) · EDA → ML → Explainability → Talk-to-Data → Docker

OBJECTIVE

What this platform does

- Predicts loan default probability for Home Credit applicants and converts it into a business-readable risk score and band (Low / Medium / High).
- Explains every prediction (SHAP) so a non-technical credit officer can see which factors drove the decision.
- Derives simple IF/THEN policy rules from the model, independent of the black-box score, for credit-policy review.
- Lets analysts ask plain-English questions about the applicant data and get validated-SQL-backed answers.
- Ships as a single Docker Compose service an evaluator can run with one command.

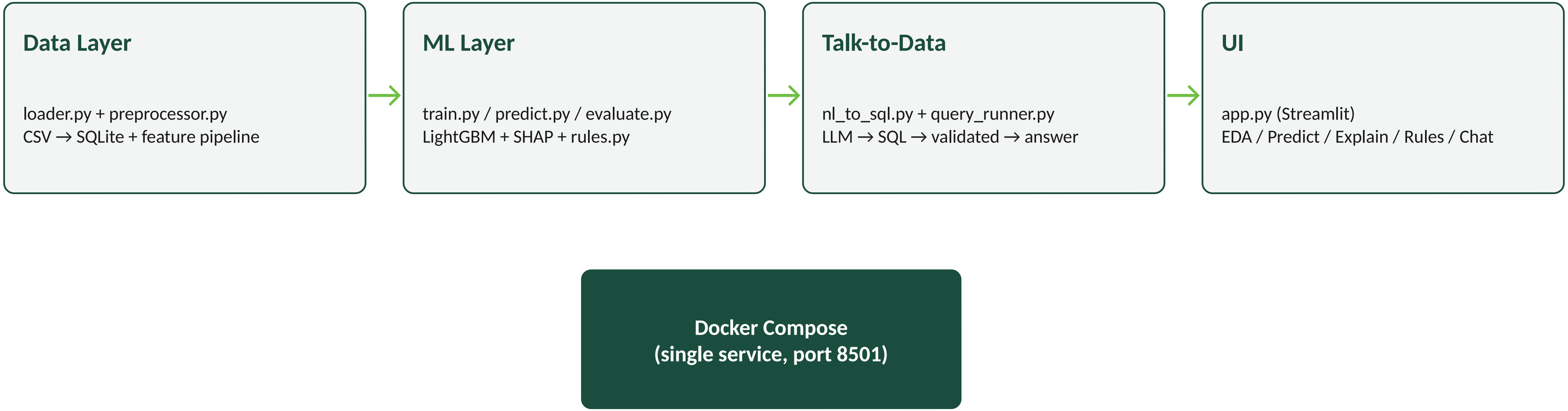
BUSINESS CONTEXT

Why this matters to a bank

- Faster decisions: automated risk scoring removes manual underwriting bottlenecks for routine applications.
- Better decisions: ROC-AUC / PR-AUC and SHAP-driven review catch risk that simple rules miss.
- Explainable & auditable: every score comes with a feature-level reason — required for regulatory review.
- Accessible: business analysts explore the applicant book in plain English, no SQL required.
- Bridges ML and policy: the rules module turns model behavior into reviewable, adjustable credit-policy language.

ARCHITECTURE

Component overview



All four components run inside one container; data/ and models/ are host-mounted volumes, never baked into the image.

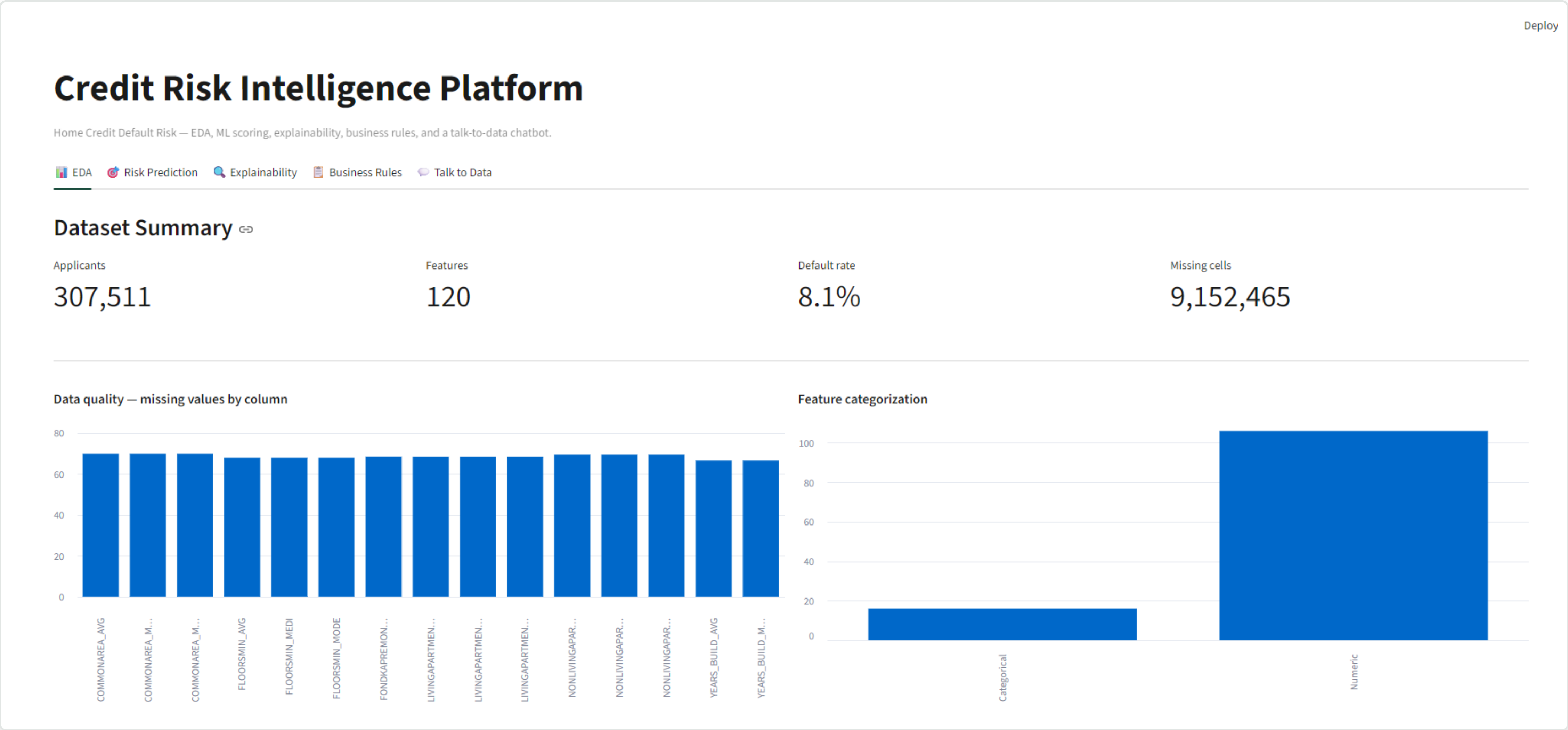
SCOPE

A deliberate scoping decision

- Home Credit ships 7 relational tables (bureau, previous applications, credit card balance, ...).
- This build works from application_train.csv — the applicant-level table — rather than joining all 7.
- Reasoning: application_train already carries demographics, income, credit terms, and several bureau-derived EXT_SOURCE scores; a full multi-table join is a multi-day feature-engineering project on its own and raises real risk of an unreliable pipeline within this assignment's time-box.
- Trade-off is explicit here and in the README — with more time, joining bureau.csv (credit-bureau history) would likely add the single highest-value signal.

EDA — LIVE FROM THE APP

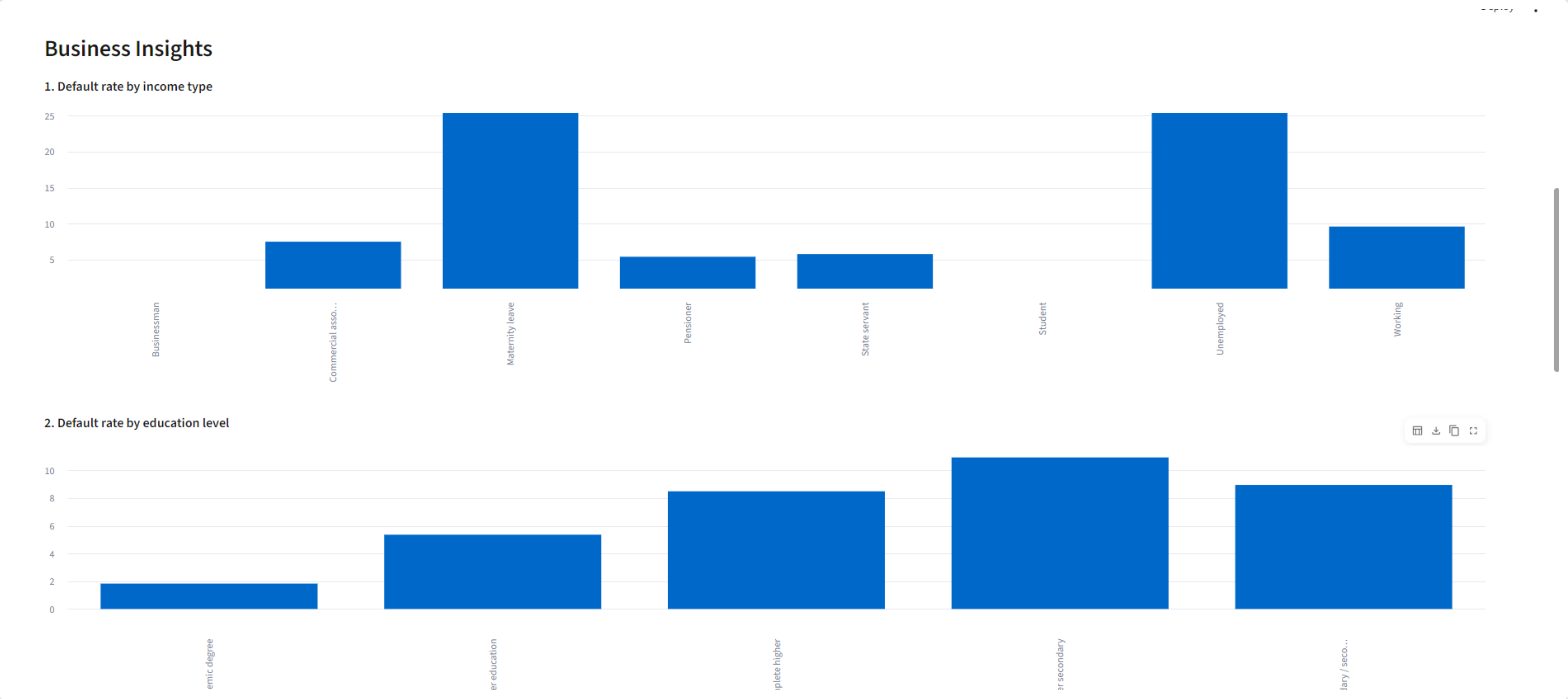
Dataset summary: the real Home Credit data



307,511 applicants · 120 features · 8.1% default rate — the full Kaggle application_train.csv, running live in the deployed app.

EDA — LIVE FROM THE APP

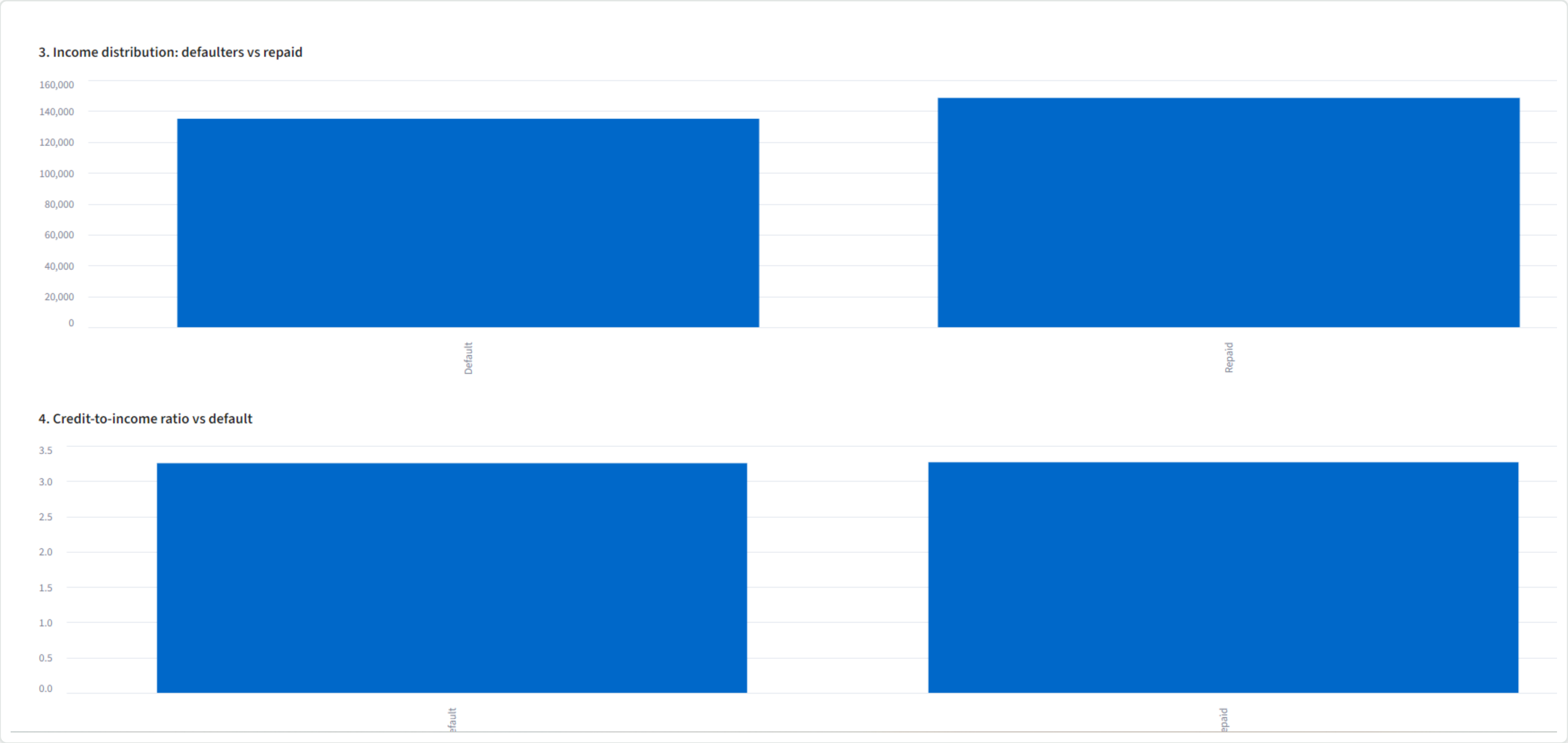
Business insight 1 & 2: income type and education



Maternity leave and Unemployed applicants show the highest default rates by income type; default rate falls steadily as education level rises.

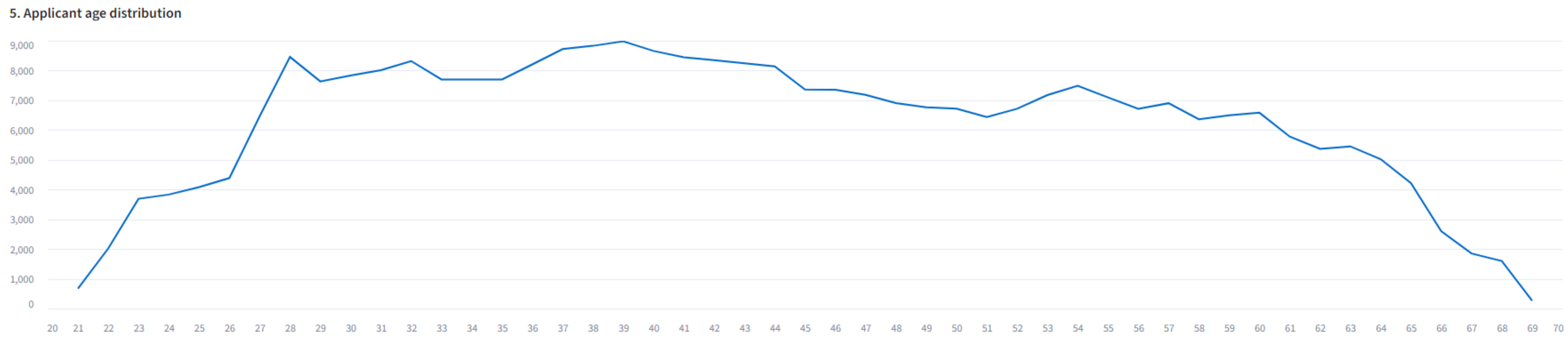
EDA — LIVE FROM THE APP

Business insight 3 & 4: income and credit-to-income ratio



Defaulters skew toward lower income and a higher credit-to-income ratio — both engineered as first-class model features (AMT_INCOME_TOTAL, CREDIT_INCOME_RATIO).

Business insight 5: applicant age, and the raw data



Raw sample data

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	NAME_TYPE_SUITE	NAME_INCOME_TYPE	NAME_EDUCATION_TY
0	100002	1	Cash loans	M	N	Y	0	202500	406597.5	24700.5	351000	Unaccompanied	Working	Secondary / secondary s
1	100003	0	Cash loans	F	N	N	0	270000	1293502.5	35698.5	1129500	Family	State servant	Higher education
2	100004	0	Revolving loans	M	Y	Y	0	67500	135000	6750	135000	Unaccompanied	Working	Secondary / secondary s
3	100006	0	Cash loans	F	N	Y	0	135000	312682.5	29686.5	297000	Unaccompanied	Working	Secondary / secondary s
4	100007	0	Cash loans	M	N	Y	0	121500	513000	21865.5	513000	Unaccompanied	Working	Secondary / secondary s
5	100008	0	Cash loans	M	N	Y	0	99000	490495.5	27517.5	454500	Spouse, partner	State servant	Secondary / secondary s
6	100009	0	Cash loans	F	Y	Y	1	171000	1560726	41301	1395000	Unaccompanied	Commercial associate	Higher education
7	100010	0	Cash loans	M	Y	Y	0	360000	1530000	42075	1530000	Unaccompanied	State servant	Higher education

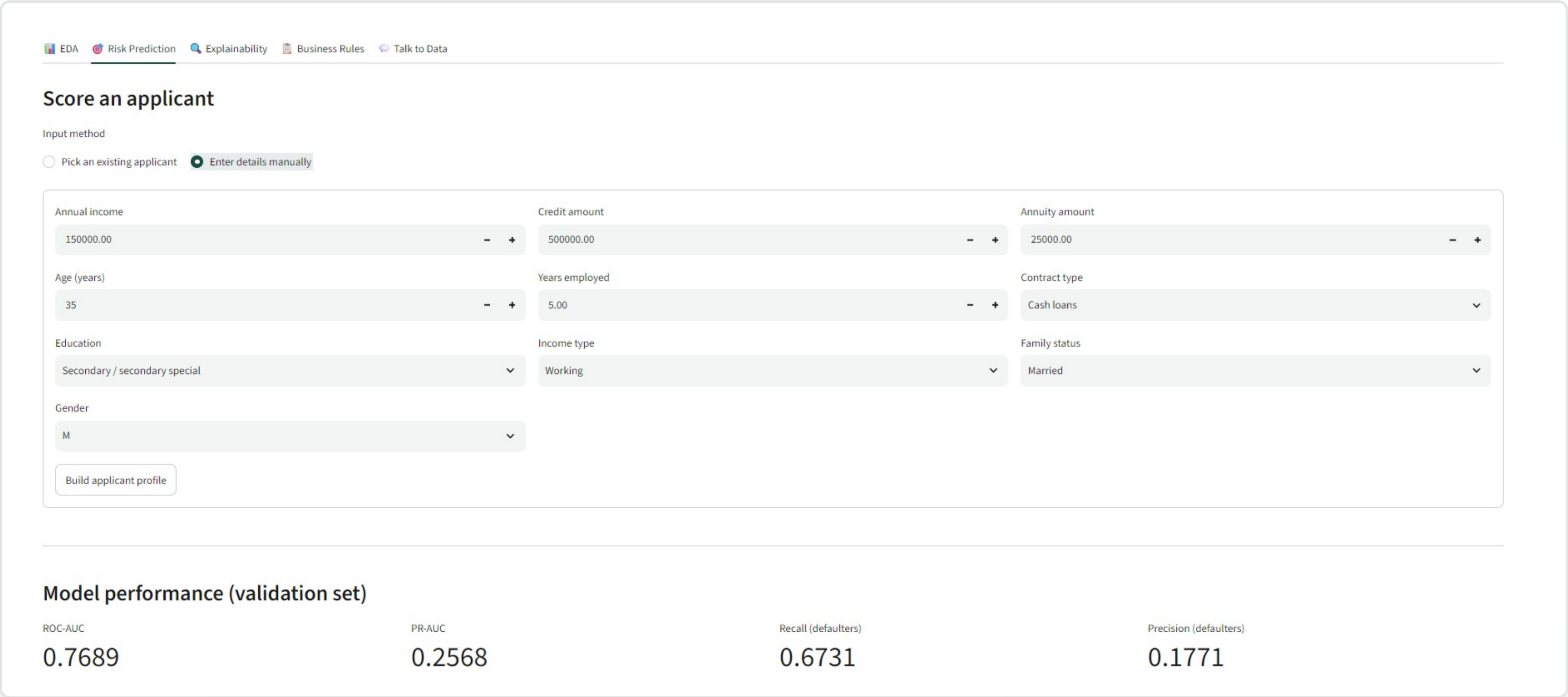
Applicant volume peaks around age 35-40 and tapers at both ends; the app also exposes the raw sample table for direct inspection.

Model design & imbalance strategy

- Model: LightGBM (LGBMClassifier) — handles mixed numeric/categorical features natively, trains in under a minute on the full 307,511-row dataset, and gives exact SHAP TreeExplainer support for Part 4.
- Imbalance handling: scale_pos_weight (cost-sensitive weighting), not SMOTE. Synthetic oversampling on high-dimensional tabular data tends to fabricate unrealistic applicant profiles without a reliable AUC gain — weighting keeps every training row real.
- Feature engineering: CREDIT_INCOME_RATIO, ANNUITY_INCOME_RATIO, CREDIT_TERM, AGE_YEARS, YEARS_EMPLOYED, EMPLOYED_AGE_RATIO — chosen to stay analyst-readable for the explainability and rules layers.
- Risk bands are calibrated as percentiles of the model's own validation output (bottom 60% = Low, next 30% = Medium, top 10% = High) rather than fixed probability cutoffs — scale_pos_weight training inflates raw probabilities, so relative score tiers give a reliable, well-populated Low/Medium/High split.

MACHINE LEARNING — LIVE FROM THE APP

Evaluation on the real 307,511-row dataset

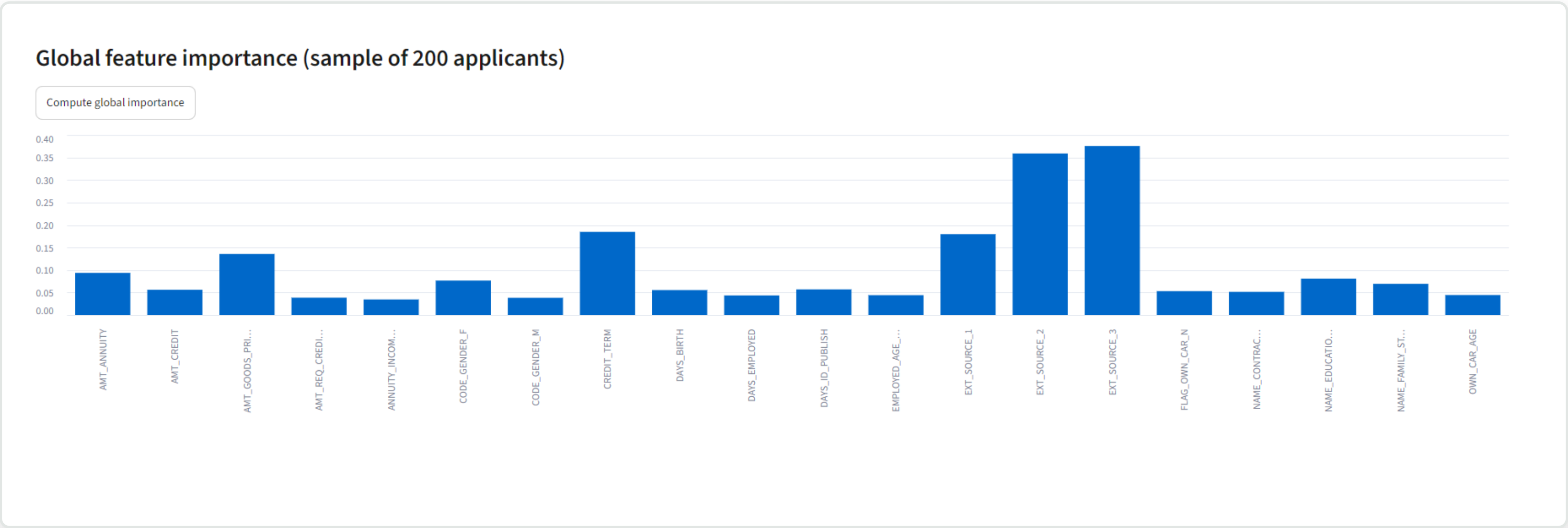


ROC-AUC and PR-AUC are the headline metrics (not accuracy) — with an 8.1% default rate, a model predicting “no default” for everyone would score >90% accuracy while being useless.

- ROC-AUC 0.7689 — solid separation between defaulters and repayers, in line with typical single-table Home Credit solutions.
- Recall (defaulters) 0.6731 — the model catches roughly two-thirds of actual defaulters.
- Precision (defaulters) 0.1771 reflects the 8% base rate: at any reasonable recall, most flagged applicants are false positives — expected and why risk banding (not a single yes/no) is the right output shape for this use case.

EXPLAINABILITY — LIVE FROM THE APP

SHAP: what drives risk across the portfolio

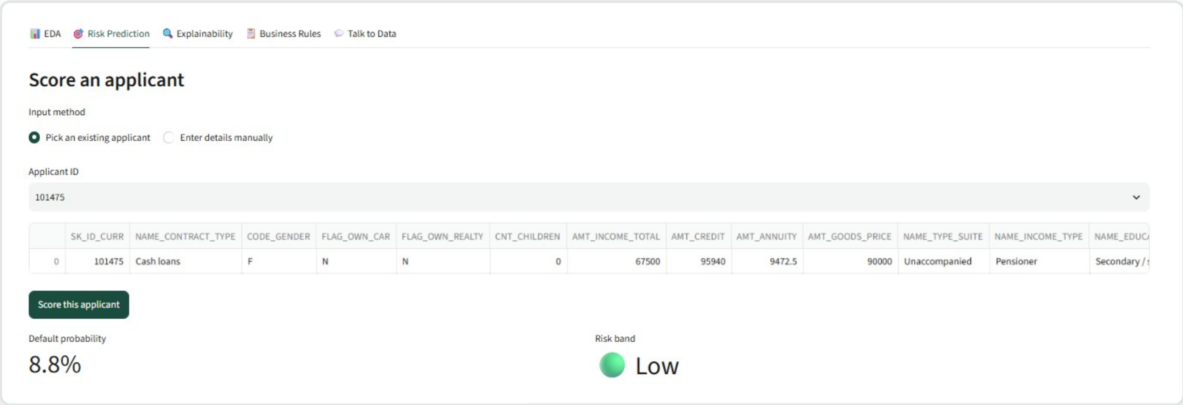


- EXT_SOURCE_2 and EXT_SOURCE_3 (bureau-derived credit scores) dominate — consistent with published Home Credit analyses.
- CREDIT_TERM (annuity/credit ratio) and EXT_SOURCE_1 are the next strongest signals.
- SHAP TreeExplainer chosen over LIME: exact for tree ensembles, fast enough for live per-applicant explanations in the UI.

Global importance computed live over a 200-applicant sample from the real dataset.

EXPLAINABILITY — LIVE FROM THE APP

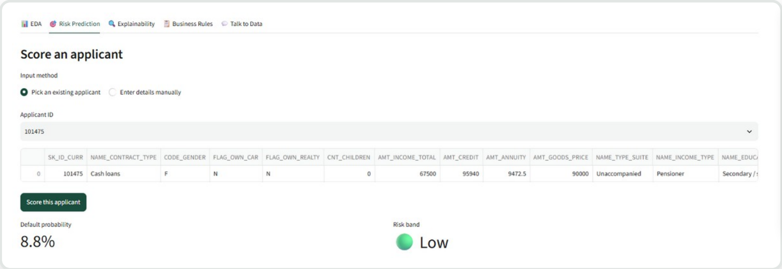
Why applicant 101475 scored Low risk (8.8%)



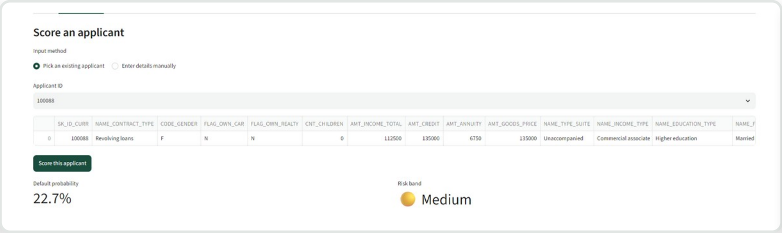
This applicant's CREDIT_TERM, EXT_SOURCE_3, and AMT_ANNUITY all pushed risk down, landing well inside the bottom 60% of scores — the same SHAP explanation the UI shows for any applicant, in real time.

RISK PREDICTION — LIVE FROM THE APP

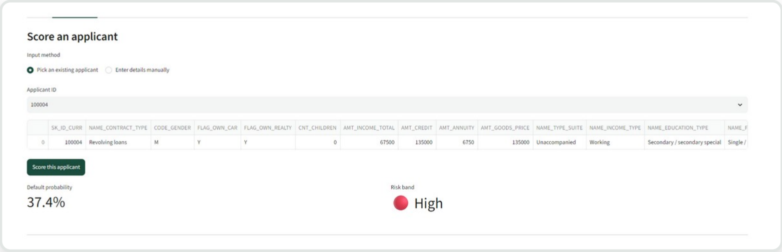
All three risk bands, scored on real applicants



Applicant 101475 — Low



Applicant 100088 — Medium

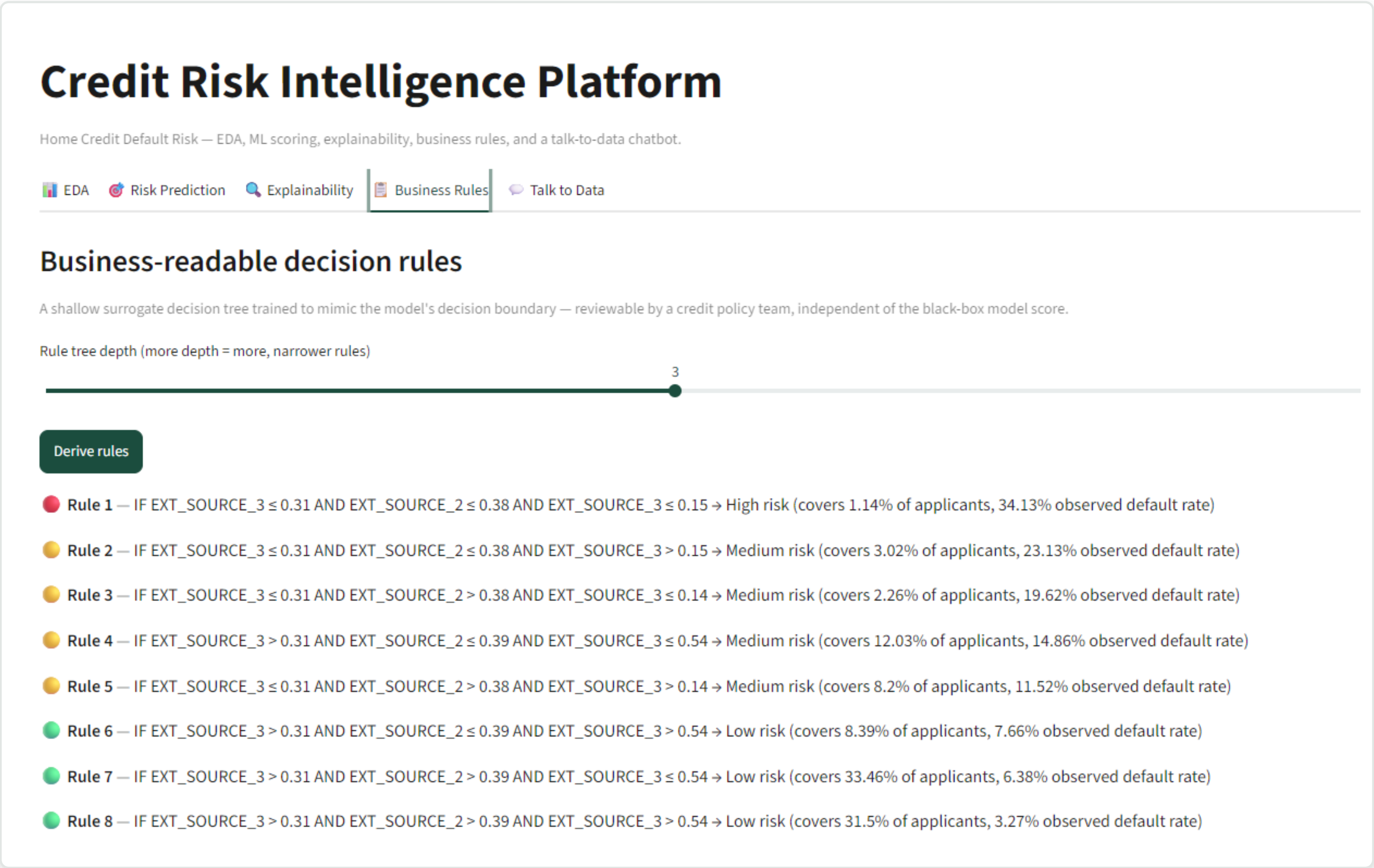


Applicant 100004 — High

Data-driven risk-band thresholds (percentiles of the model's own score distribution) reliably produce all three tiers — not just Medium/High — across real applicants from the dataset.

BUSINESS RULES — LIVE FROM THE APP

From black-box score to reviewable policy

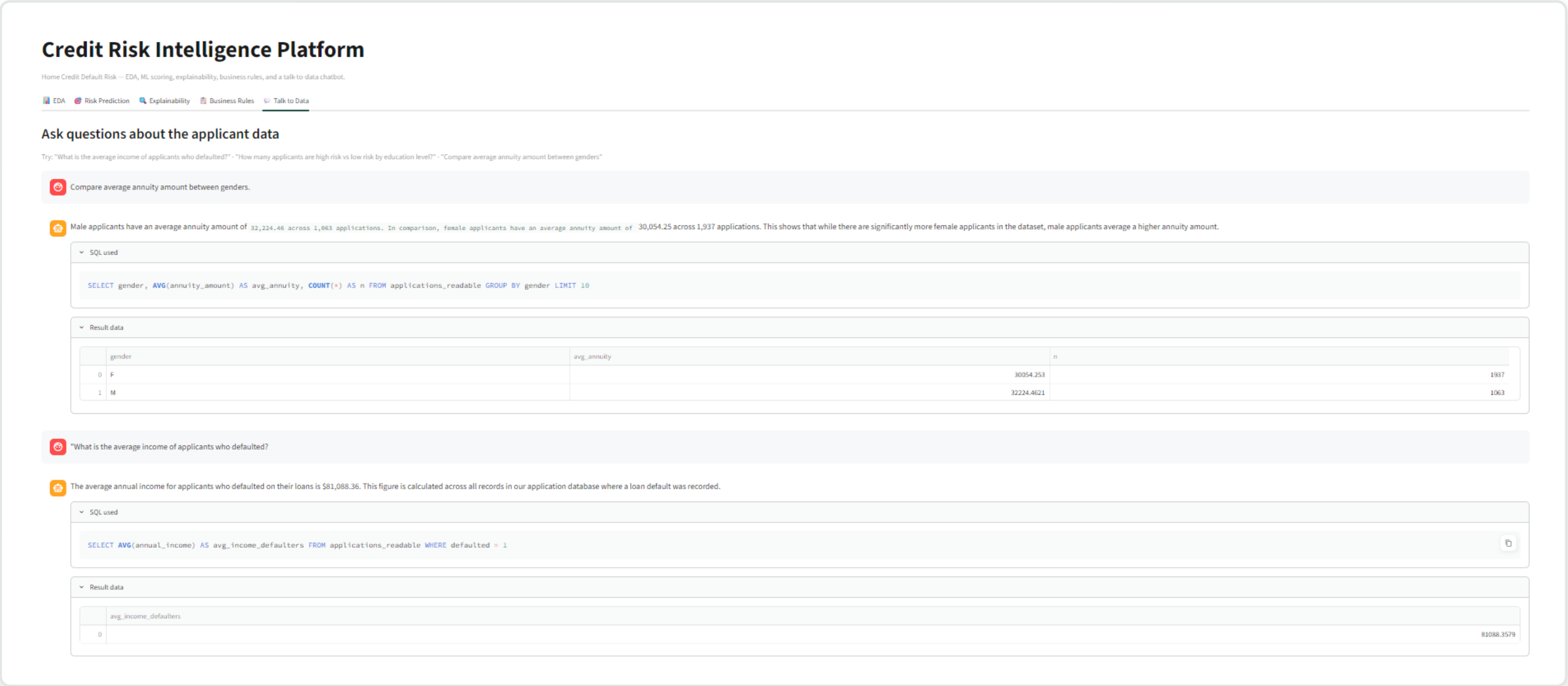


All 8 rules derived live from the real dataset (depth-3 surrogate tree) — EXT_SOURCE_2 and EXT_SOURCE_3 anchor every split, matching the SHAP global-importance ranking.

- Approach: a shallow surrogate decision tree mimics the production model's decision boundary in plain IF/THEN language a credit-policy team can review directly.
- Rule 1 (highest risk): EXT_SOURCE_3 ≤ 0.31 AND EXT_SOURCE_2 ≤ 0.38 AND EXT_SOURCE_3 ≤ 0.15 → 34.13% observed default rate.
- Rule 8 (lowest risk): EXT_SOURCE_3 > 0.31 AND EXT_SOURCE_2 > 0.39 AND EXT_SOURCE_3 > 0.54 → 3.27% observed default rate, covering 31.5% of applicants.

TALK-TO-DATA — LIVE FROM THE APP

Natural language, validated SQL, real answers



Two real exchanges against the full 307,511-row dataset, powered by Gemini (free tier) — SQL is shown and validated before execution, and every number in the answer comes from the actual query result.

- “Compare average annuity between genders” → correct SQL, correct grouped result, grounded plain-English answer.
- “Average income of applicants who defaulted” → $81,088.36, computed directly from the real data, not estimated.
- SQL safety: SELECT-only, single-statement, allow-listed tables, forbidden-keyword check, automatic LIMIT — verified against DROP/UPDATE/multi-statement attempts.

Prompt engineering, token optimization & provider flexibility

- Two-stage LLM flow: question → SQL (validated before execution) → real query results → plain-English answer. The answer stage only ever sees data that actually came back from SQLite — it cannot invent numbers.
- Schema exposed to the LLM is a trimmed, business-readable view (13 columns) rather than the raw 122-column table — the core token-optimization decision.
- Multi-provider by design: Anthropic, OpenAI, or Gemini (free tier, no billing) — swap providers via one .env variable, no code change.
- Gemini-specific hardening: automatic detection and one-retry-with-larger-budget when a response is truncated mid-generation (finish_reason=MAX_TOKENS), so a tight token budget never silently returns a broken SQL query or a cut-off answer.
- Conversation memory: the last 4 turns are passed back to the LLM so follow-up questions resolve correctly, without unbounded token growth.

Docker deployment, limitations, and next steps

- docker-compose up --build builds the single Streamlit service, mounts ./data and ./models as volumes (dataset and model artifacts are never baked into the image or committed to git), and exposes the app on :8501 — verified working end-to-end on the full real dataset.
- Startup is self-diagnosing: the UI detects a missing dataset or an untrained model and tells the evaluator exactly what to do, with a one-click “Train model” fallback in-app.
- Known limitations: scoped to the applicant-level table only (see slide 5); raw model probabilities are uncalibrated due to scale_pos_weight training, addressed via percentile-based risk bands rather than a separate calibration model.
- Next steps with more time: join bureau.csv / previous_application.csv for stronger signal; add proper probability calibration (CalibratedClassifierCV); cache SHAP results; add authentication in front of the chatbot.